

The Model

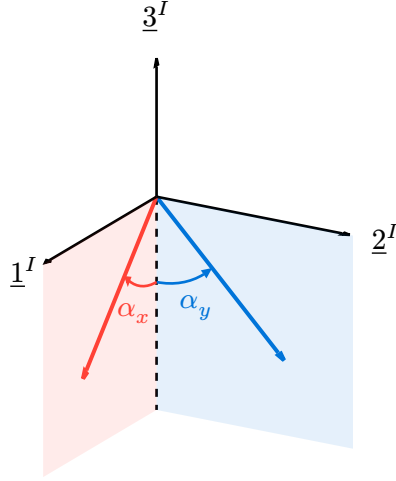
$$m_P \underline{a}_P = T \underline{q} - m_P \underline{g}$$

$$m_D \underline{a}_D = \underline{f} - T \underline{q} - m_D \underline{g}$$

$$\frac{d\underline{\omega}^{BE}}{dt} = (\underline{J}_B^B)^{-1} [\underline{n}_B - \underline{\Omega}^{BE} \underline{J}_B^B \underline{\omega}^{BE}]$$

$$\underline{s}_P = \underline{s}_D - L \underline{q}$$

swing angles



EKF

Frames

$[\cdot]^I$: ENU (+x East, +y North, +z Up)

$[\cdot]^B$: body (+x starboard, +y nose, +z up)

$[\cdot]^C$: camera (+x right, +y down, +z optical axis)

$$\Pi = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

State

$$[\xi]^I = \begin{pmatrix} \alpha_x \\ \alpha_y \\ \dot{\alpha}_x \\ \dot{\alpha}_y \\ \psi_P \end{pmatrix} \in \mathbb{R}^5$$

$$s_x = \sin \alpha_x, \quad c_x = \cos \alpha_x, \quad s_y = \sin \alpha_y, \quad c_y = \cos \alpha_y$$

α_x, α_y : tether swing angles, tipping toward $+x^I$ (East) and $+y^I$ (North) [rad]

$\dot{\alpha}_x, \dot{\alpha}_y$: swing rates [rad/s]

ψ_P : payload yaw: marker-row direction, from $+x^I$ toward $+y^I$ [rad]

Input

$$[T]^{IB} = \begin{pmatrix} c_\theta c_\psi & s_\varphi s_\theta c_\psi - c_\varphi s_\psi & c_\varphi s_\theta c_\psi + s_\varphi s_\psi \\ c_\theta s_\psi & s_\varphi s_\theta s_\psi + c_\varphi c_\psi & c_\varphi s_\theta s_\psi - s_\varphi c_\psi \\ -s_\theta & s_\varphi c_\theta & c_\varphi c_\theta \end{pmatrix}$$

$$[C]^{IB} = \Pi [T]^{IB} \Pi$$

$$[a_s]^I = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = [C]^{IB} \Pi \left(\frac{g}{1000} [a_s]_{\text{NED}}^B \right), \quad [a_s]_{\text{hover}}^I = \begin{pmatrix} 0 \\ 0 \\ g \end{pmatrix}$$

$[a_s]^I$: specific force

Process

$$[\dot{\xi}]^I = f(\xi, a_s) = \begin{pmatrix} \dot{\alpha}_x \\ \dot{\alpha}_y \\ -\frac{c_x a_1 + s_x a_3}{L c_y} + 2 \frac{s_y}{c_y} \dot{\alpha}_x \dot{\alpha}_y \\ -\frac{c_y a_2 + s_y (c_x a_3 - s_x a_1)}{L} - s_y c_y \dot{\alpha}_x^2 \\ 0 \end{pmatrix}$$

$$F = \frac{\partial f}{\partial \xi} = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ \frac{s_x a_1 - c_x a_3}{L c_y} & -\frac{(c_x a_1 + s_x a_3) s_y}{L c_y^2} + \frac{2 \dot{\alpha}_x \dot{\alpha}_y}{c_y^2} & \frac{2 s_y}{c_y} \dot{\alpha}_y & \frac{2 s_y}{c_y} \dot{\alpha}_x & 0 \\ \frac{s_y (c_x a_1 + s_x a_3)}{L} & \frac{s_y a_2 - c_y (c_x a_3 - s_x a_1)}{L} - \cos(2 \alpha_y) \dot{\alpha}_x^2 & -2 s_y c_y \dot{\alpha}_x & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

L : pivot to payload CG [m]

$F \in \mathbb{R}^{5 \times 5}$: process Jacobian (continuous-time)

Predict

Converting Jacobian to discrete-time w/ RK4

$$k_1 = f(\hat{\xi}_k^+, a_s), \quad k_2 = f\left(\hat{\xi}_k^+ + \frac{\Delta t}{2} k_1, a_s\right)$$

$$k_3 = f\left(\hat{\xi}_k^+ + \frac{\Delta t}{2} k_2, a_s\right), \quad k_4 = f(\hat{\xi}_k^+ + \Delta t k_3, a_s)$$

$$\hat{\xi}_{k+1}^- = \hat{\xi}_k^+ + \frac{\Delta t}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$\Phi_k = I_5 + F \Delta t + \frac{1}{2} (F \Delta t)^2, \quad Q = \text{diag}(0, 0, q, q, q_\psi) \Delta t$$

$$P_{k+1}^- = \Phi_k P_k^+ \Phi_k^\top + Q$$

Δt : time step [s]

$\hat{\xi}_k^-, \hat{\xi}_k^+$: estimate before & after the update at step k

$P_k^-, P_k^+ \in \mathbb{R}^{5 \times 5}$: covariance before & after

Φ_k : discrete time Jacobian

q : process-noise intensity

Geometry

$$[\hat{n}]^I = \begin{pmatrix} s_x c_y \\ s_y \\ -c_x c_y \end{pmatrix}, \quad [\hat{m}]^I = \begin{pmatrix} \cos \psi_P \\ \sin \psi_P \\ 0 \end{pmatrix}, \quad \text{assuming no roll, pitch}$$

$[\hat{n}]^I$: unit vector, tether pivot $\rightarrow$ payload

$[\hat{m}]^I$: unit vector along the marker row

Measurement, $j \in V_k$

$$[p_j]^I = L_m [\hat{n}]^I - o_j [\hat{m}]^I$$

V_k : marker IDs detected in frame k

L_m : pivot to marker-board center

o_j : signed marker offset from center along the row, +left

$[p_j]^I$: position of marker j

$$[p_j]^C = \begin{pmatrix} X_j \\ Y_j \\ Z_j \end{pmatrix} = [C]^{CB} \left([\ell]^B + [C]^{BI} [p_j]^I - [t_{BC}]^B \right)$$

$[t_{BC}]^B$: camera optical center (offset due to science stick) [m]

$[\ell]^B$: tether pivot point offset [m]

$$h_j(\xi) = \begin{pmatrix} f_u X_j / Z_j + u_0 \\ f_v Y_j / Z_j + v_0 \end{pmatrix}$$

f_u, f_v, u_0, v_0 : intrinsics [px]

h_j : predicted pixel in image plane

$$H_j = \frac{\partial h_j(\xi)}{\partial [p_j]^C} \frac{\partial [p_j]^C}{\partial [p_j]^I} \frac{\partial [p_j]^I}{\partial [\xi]^I}$$

$$H_j = \begin{pmatrix} \frac{f_u}{Z_j} & 0 & -\frac{f_u X_j}{Z_j^2} \\ 0 & \frac{f_v}{Z_j} & -\frac{f_v Y_j}{Z_j^2} \end{pmatrix} [C]^{CB} [C]^{BI} \left(L_m \frac{\partial [\hat{n}]^I}{\partial \xi} - o_j \frac{\partial [\hat{m}]^I}{\partial \xi} \right)$$

$H_j \in \mathbb{R}^{2 \times 5}$: Jacobian of h_j w.r.t. ξ , by chain rule

$[\ell]^B, [t_{BC}]^B$ are constant

Update

$$z^{(j)} = h_j(\xi) + \varepsilon^{(j)}$$

$m_k \in \{0, 1, 2, 3\}$: number of markers spotted in a given frame

$z^{(j)}$: pixel logged for marker j [px]

$\varepsilon^{(j)}$: measurement noise

$$z_k = \begin{pmatrix} z^{(1)} \\ \vdots \\ z^{(m_k)} \end{pmatrix}, \quad h_k = \begin{pmatrix} h_1 \\ \vdots \\ h_{m_k} \end{pmatrix}, \quad H_k = \begin{pmatrix} H_1 \\ \vdots \\ H_{m_k} \end{pmatrix}, \quad R_k = (\sigma_{\text{det}}^2 + f_u^2 \sigma_\theta^2) I_{2m_k}$$

$$z_k = h_k(\xi) + \varepsilon_k, \quad \varepsilon_k \sim \mathcal{N}(0, R_k)$$

$$y_k = z_k - h_k, \quad S_k = H_k P_k^- H_k^\top + R_k, \quad K_k = P_k^- H_k^\top S_k^{-1}$$

$$\hat{\xi}_k^+ = \hat{\xi}_k^- + K_k y_k$$

$$P_k^+ = (I_5 - K_k H_k) P_k^-$$

y_k : innovation S_k : innovation covariance $K_k \in \mathbb{R}^{5 \times 2m_k}$: Kalman gain
 σ_{det} : marker detection noise [px] σ_θ : attitude 1σ [rad]

If no markers detected:

$$m_k = 0 \implies \hat{\xi}_k^+ = \hat{\xi}_k^-, \quad P_k^+ = P_k^-$$